HARIHARAN J

EMBEDDED LINUX & FIRMWARE ENGINEER

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PROFESSIONAL SUMMARY

Embedded Linux & Firmware Engineer with 1+ year of experience developing production firmware for Linux-based Ethernet camera systems and embedded multimedia devices. Experienced in Embedded C/C++, Embedded Linux, system programming, networking, multimedia frameworks, and real-time streaming applications. Skilled in ONVIF implementation, GStreamer, socket programming, secure firmware updates, and low-level Linux debugging. Seeking opportunities in Embedded Linux, Firmware, BSP, and low-level system software development.

PROFESSIONAL EXPERIENCE

Project Engineer (Camera Products), e-con Systems India Pvt. Ltd., Chennai Aug 2024 - Present

- Developed production firmware for Linux-based Ethernet camera systems using Embedded C/C++.
- Implemented ONVIF Profiles **S, T, G and M**, enabling device discovery, media streaming, recording, event management, PTZ control, metadata, and interoperability with third-party ONVIF-compliant clients.
- Developed multimedia streaming pipelines using **GStreamer**, **RTSP, RTP, H.264/H.265, V4L2**, and ALSA for embedded Linux platforms.
- Designed and implemented multi-camera synchronization using **IEEE 1588 Precision Time Protocol (PTP)** for accurate frame synchronization across Ethernet-connected cameras.
- Developed secure OTA firmware update mechanisms including firmware package verification, RSA signature validation, firmware version verification, and secure installation.
- Implemented networking features using **Ethernet**, **TCP/IP, UDP, HTTP, HTTPS, RTSP**, and socket programming.
- Investigated and resolved complex system-level issues involving Linux kernel subsystems, networking, multimedia frameworks, memory management, DMA buffers, and camera SDK integration.
- Optimized firmware performance, streaming latency, memory utilization, and system stability for production deployments.
- Collaborated with hardware, validation, and application teams throughout firmware development, debugging, integration, and customer issue resolution.

PROFESSIONAL PROJECTS

1. ONVIF Device Server Implementation

Technologies: Embedded C, ONVIF, SOAP, HTTP, RTSP

- Implemented ONVIF Profiles S, T, G and M for embedded IP camera products.
- Developed device discovery, media service, PTZ, recording management, event handling, and metadata support.
- Improved interoperability with third-party Video Management Systems (VMS) and industrial vision platforms.
- Ensured protocol compliance and reliable operation in production firmware.

2. Multi-Camera Synchronization using Precision Time Protocol (PTP)

Technologies: IEEE1588, Ethernet, Embedded Linux

- Developed synchronization logic for multiple Ethernet-connected cameras.
- Implemented precise timestamp synchronization and frame alignment.
- Improved synchronization accuracy for machine vision and industrial imaging applications.
- Optimized firmware for deterministic real-time performance.

3. Secure OTA Firmware Update System

Technologies: C, OpenSSL, Bash, Linux

- Designed a secure firmware update workflow for embedded Linux devices.
- Implemented RSA-based firmware package signature verification.

- Added firmware version validation to prevent rollback and unauthorized firmware installation.
- Improved firmware reliability and deployment security.

4. Embedded Multimedia Streaming Framework

Technologies: GStreamer, RTSP, RTP, H.264, H.265, ALSA, V4L2

- Developed multimedia streaming pipelines for embedded IP cameras.
- Integrated video encoding, audio capture, and real-time network streaming.
- Investigated synchronization, latency, and multimedia buffer management issues.
- Improved streaming performance and system stability.

5. Embedded Firmware Diagnostics Framework

Technologies: Embedded Linux, C, Bash

- Developed diagnostic utilities for autofocus, streaming, networking, and firmware health monitoring.
- Investigated Linux kernel logs, memory allocation failures, multimedia buffer management, and system resource utilization.
- Performed root cause analysis of production firmware issues involving Linux kernel, networking, and multimedia subsystems.
- Improved firmware robustness through systematic debugging and performance optimization.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python and Bash.
- **Embedded Systems:** Embedded Linux, POSIX APIs, Multithreading, Cross Compilation, Yocto, Buildroot, Firmware Development and Linux System Programming.
- **Networking:** TCP/IP, UDP, HTTP, HTTPS, RTSP, RTP, ONVIF, Socket Programming, Ethernet, DHCP, IPv4, IPv6, PTP (IEEE1588).
- **Multimedia:** GStreamer, V4L2, ALSA, H.264, H.265, MJPEG, Video Streaming.
- **Tools:** Git, Make, CMake, GDB, Linux Shell Scripting, OpenSSL.

EDUCATION

B.E - Electrical and Electronics Engineering 2020 – 2024

Kumaraguru College of Technology, Coimbatore.

CGPA – 9.49

ACHIEVEMENTS

- **Extra Mile Award** – e-con System India Pvt. LTD. (2025).
- **Best Outgoing Student Award –** Department of Electrical and Electronics Engineering, Kumaraguru College of Technology (2024).
- **Runner-Up -** Fit.Kovai Hackathon**,** Bosch Global Software Technologies (2023).
- **Mahatma Gandhi Merit Scholarship Award –** Awarded four consecutive years for academic excellence from Kumaraguru College of Technology (2021 - 2024).